

ORF309/MAT380

**Bernoulli Processes I: Definitions
& Binomial Distributions**

(pp.37-42)

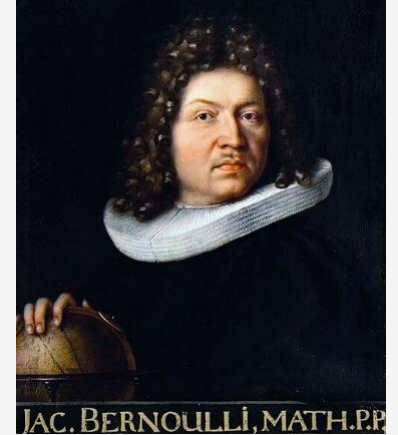
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Agenda for today

- *Bernoulli DRVs and Processes*
- *Binomial distribution*
- Counting Successes (& Failures)
- Expected number of successes (& failures)
- Successes between trials
- Incremental successes

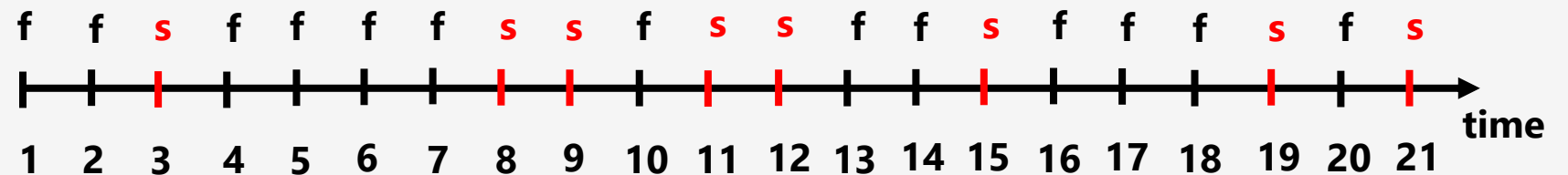
Definitions

- For the next 3-4 lectures we will discuss Bernoulli Processes
 - Named after Jacob Bernoulli (1654-1705)
- **Definition:** Any random variable $X: \Omega \rightarrow \{0, 1\}$ is called **Bernoulli RV**.
 - The **indicator function** is a Bernoulli variable (1_A is a Bernoulli RV, for every event A)
- **Definition:** A **Bernoulli Process** is an (infinite) **sequence** $X_1, X_2, X_3, \dots$ of **i**ndependent **i**dentically **d**istributed (**i.i.d.**) **Bernoulli RVs**, with $P\{X_i = 1\} = p$ and $P\{X_i = 0\} = 1 - p$
- **Remark:** We will define $1 - p = q$



Interpretation and Examples of Bernoulli Processes

- A Bernoulli Process **counts Successes** ($\{X_i = 1\}$) and **Failures** ($\{X_i = 0\}$)



- Examples:**
 - An assembly line where a specific device is produced. A **Success** means the **device can be sold to customers**; and a **Failure** means the **device malfunctions**.
 - Asking random people whether they support a specific policy (yes/no answer)
 - Whether I get a ticket if I park my car at a specific spot in Princeton

Questions we can answer using Bernoulli Processes!

- **Important Questions:**

1. After n trials, how many successes do we have?
2. How long do we need to wait for the 1st (or 2nd, 3rd, ...) success (or failure)?
3. Will a success (or failure) eventually show up?
4. Can we fail (or succeed) for ever?

Counting Successes (or Failures)

- **Remember:** Bernoulli variables can be used to count successes
- $S_n = X_1 + X_2 + X_3 + \cdots + X_n$ is a **new random variable** that counts the number of successes in n trials (or by time n)
- X_i is a Bernoulli RV ($i = 1, 2, 3, \dots, n$)
 - $X_i \in \{0, 1\}$ and $P\{X_i = 1\} = p$, $P\{X_i = 0\} = q$, $p + q = 1$

Counting Successes (or Failures)

- **Example:**
 - Let's compute the **probability** of the event: "exactly 2 successes come up in the first 8 trials."
 - How can we express this event in terms of the RV: S_n ??

Counting Successes (or Failures)

- **Example:**
 - Let's compute the **probability** of the event: "exactly 2 successes come up in the first 8 trials."
 - How can we express this event in terms of the RV: $S_n \rightarrow \{S_8 = 2\}$
 - We are looking to compute the probability: $P\{S_8 = 2\} = ?$
 - **Possible realizations** of the event $\{S_8 = 2\}$:
 - $\{S_8 = 2\} = \{ssfffff, sf sfffff, sff sffff, \dots, ffffffss\}$

- **Example (cont.)**

- Let's now compute: $P\{\textcolor{red}{s}sffff\} = ?$

- **Example (cont.)**

- Let's now compute: $P\{\textcolor{red}{s}sffff\} = ?$

- Each trial is **independent** so we can write:

$$\begin{aligned} P\{\textcolor{red}{s}sffff\} &= P\{X_1 = 1, X_2 = 1, X_3 = 0, \dots, X_8 = 0\} \stackrel{\boxed{||}}{=} \\ &= P\{X_1 = 1\}P\{X_2 = 1\}P\{X_3 = 0\} \dots P\{X_8 = 0\} = \textcolor{red}{p}pqq \dots q = p^2q^6 \end{aligned}$$

- **Example (cont.)**

- Let's now compute: $P\{\text{ssffff}\} = ?$

- Each trial is **independent** so we can write:

$$\begin{aligned} P\{\text{ssffff}\} &= P\{X_1 = 1, X_2 = 1, X_3 = 0, \dots, X_8 = 0\} = \boxed{\text{||}} \\ &= P\{X_1 = 1\}P\{X_2 = 1\}P\{X_3 = 0\} \dots P\{X_8 = 0\} = \underset{p}{p}\underset{p}{p}\underset{q}{q}q \dots q = p^2q^6 \end{aligned}$$

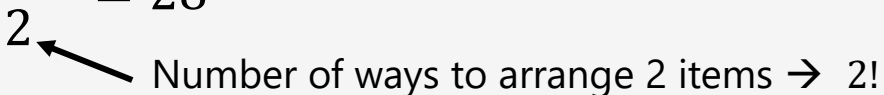
- Working similarly, we can compute all the other probabilities, which are all equal to p^2q^6
- Hence: $P\{S_8 = 2\} = p^2q^6 \times (\#\{\text{arrangements of 2 successes in 8 trials}\})$

- **Example (cont.)**

- ($\#\{\text{arrangements of 2 successes in 8 trials}\}$) is equivalent to
 - # of ways to put 2 **identical** balls in 8 bins or
of ways to chose 2 slots out of 8 slots

- **Example (cont.)**

- ($\#\{\text{arrangements of 2 successes in 8 trials}\}$) is equivalent to
- # of ways to put 2 **identical** balls in 8 bins or
of ways to chose 2 slots out of 8 slots

This is equal to $\frac{8 \times 7}{2} = 28$
 Number of ways to arrange 2 items $\rightarrow 2!$

or equivalently:
$$\binom{8}{2} = \frac{8!}{2!6!} = \frac{1 \times 2 \times \dots \times 7 \times 8}{1 \times 2 \times 1 \times 2 \times \dots \times 6} = \frac{7 \times 8}{2}$$

- **Example (cont.):**

- **Generalization:** we can compute the probability to have **exactly k successes** in **n trials** $P\{S_n = k\} = ?$ where $1 \leq k \leq n$

- $P\{S_n = k\} = p^k q^{n-k} \times (\#\{\text{arrangements of } k \text{ successes in } n \text{ trials}\})$

$$(\#\{\text{arrangements of } k \text{ successes in } n \text{ trials}\}) = \frac{n(n-1) \dots (n-k+1)}{k(k-1) \dots 1} = \binom{n}{k}$$

$$P\{S_n = k\} = \binom{n}{k} p^k q^{n-k}$$

Number of ways to arrange k items $\rightarrow k!$

Binomial coefficient;
Pronounced: " n chose k "

Binomial Distribution

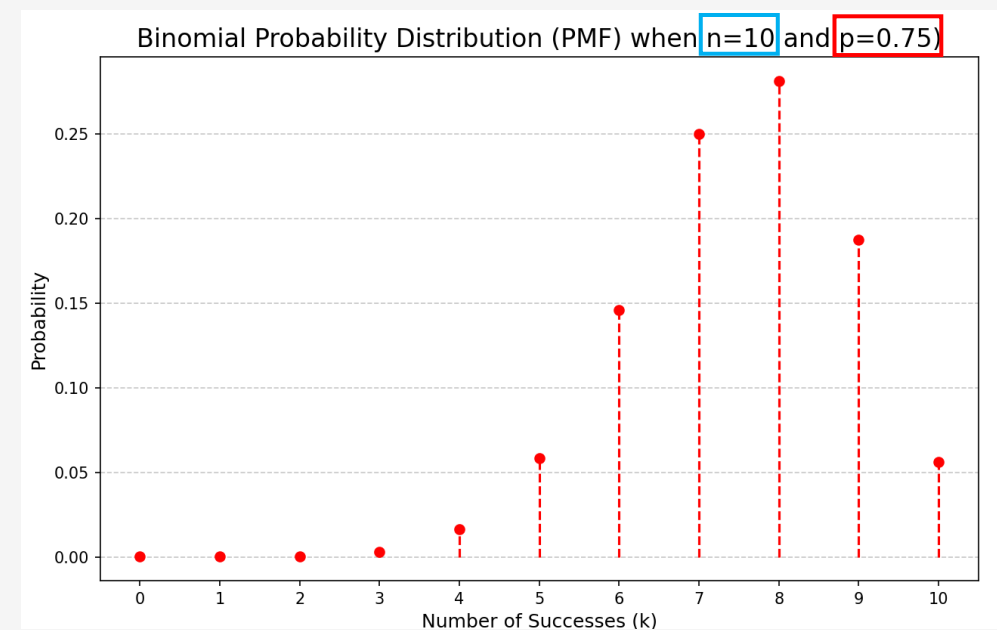
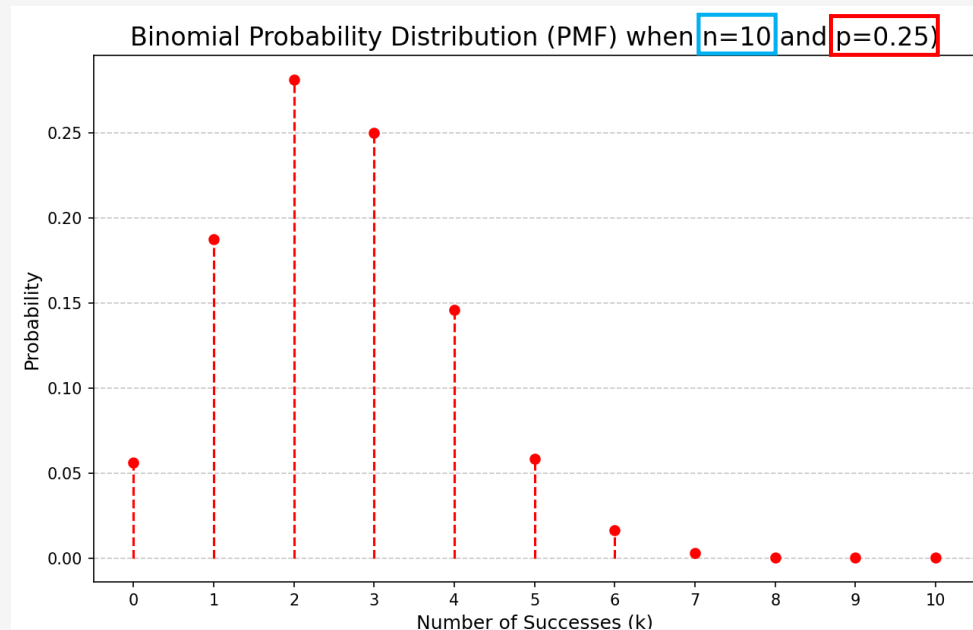
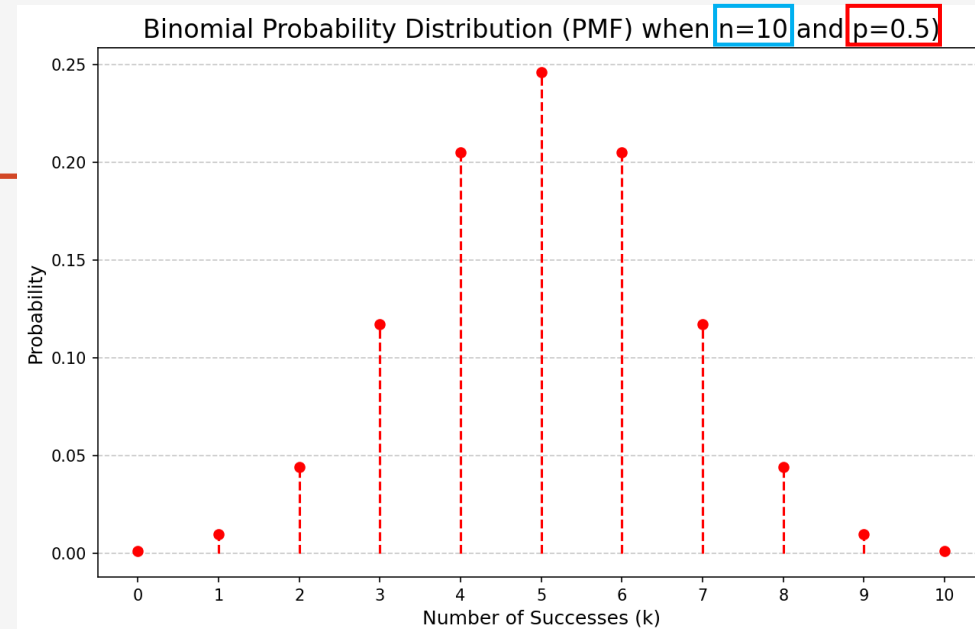
- **Definition:** A random variable Z with values $\{0,1,2, \dots, n\}$ is said to have the **Binomial Distributions** if

$$P\{Z = k\} = \binom{n}{k} p^k q^{n-k}$$

- where, $k \in \{0,1,2, \dots, n\}$ and $p \in [0,1]$, $q = 1 - p$.
- **Notation:** $Z \sim \text{Binom}(n, p)$
- **Remark:** $S_n \sim \text{Binom}(n, p)$ and $S_{n+1} \sim \text{Binom}(n + 1, p)$

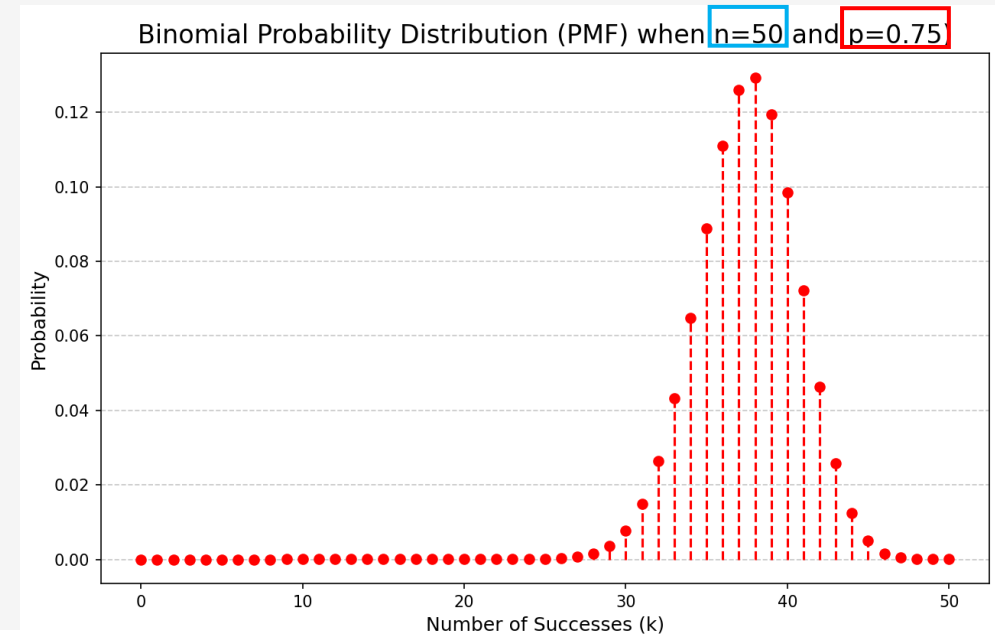
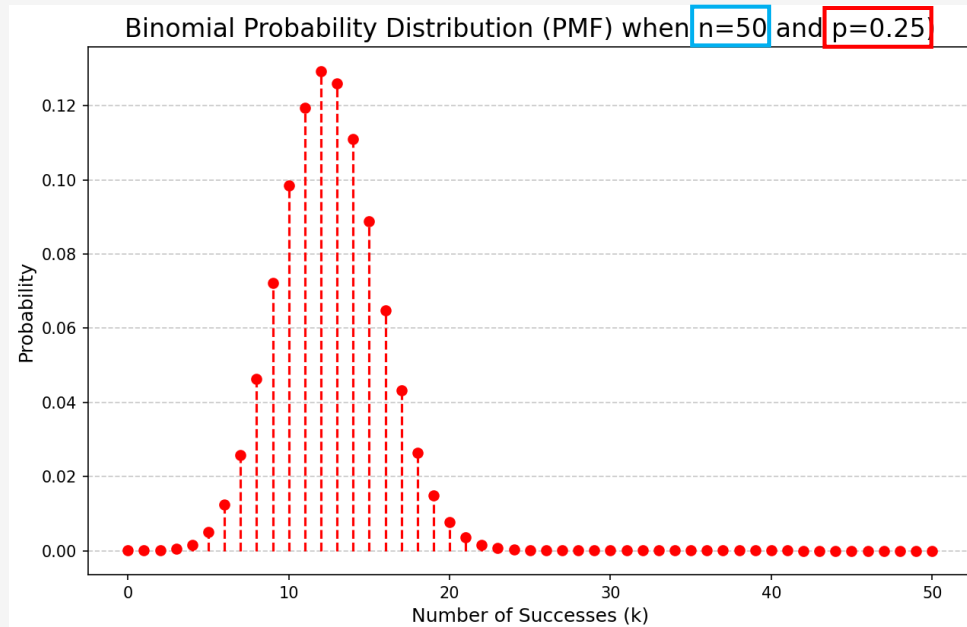
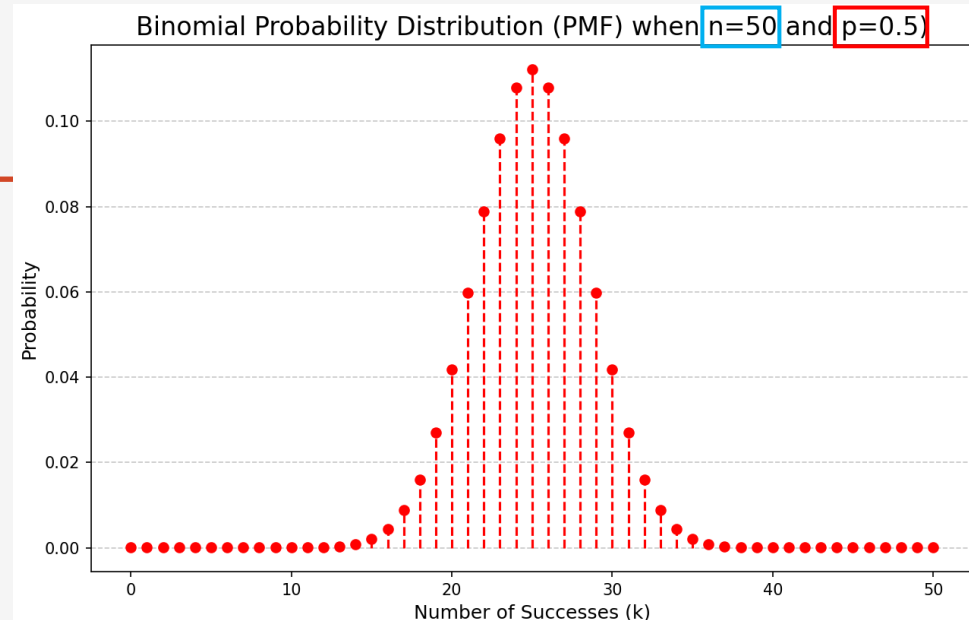
$n = 10$

$$P\{Z = k\} = \binom{n}{k} p^k q^{n-k}$$



$n = 50$

$$P\{Z = k\} = \binom{n}{k} p^k q^{n-k}$$



Binomial Distribution

- **Sanity Check:**

- Let's check if $\sum_{k=0}^n P\{Z = k\} = 1$

$$\sum_{k=0}^n P\{Z = k\} = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k} =$$

What does this remind you??

Binomial Distribution

- **Sanity Check:**

- Let's check if $\sum_{k=0}^n P\{Z = k\} = 1$

$$\sum_{k=0}^n P\{Z = k\} = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k} = (p + q)^n = (p + 1 - p)^n = 1$$

Expected number of successes

- **Question:** What is the **average** #successes in n trials ($E[S_n]$) ??

$$E[S_n] = \sum_{k=0}^n k P\{S_n = k\} = \sum_{\substack{k=1 \\ \text{red}}}^n k \binom{n}{k} p^k q^{n-k} \quad (1)$$

- Let's compute:

$$k \binom{n}{k} =$$

Expected number of successes

- **Question:** What is the **average** #successes in n trials ($E[S_n]$) ??

$$E[S_n] = \sum_{k=0}^n k P\{S_n = k\} = \sum_{\substack{k=1 \\ \text{red}}}^n k \binom{n}{k} p^k q^{n-k} \quad (1)$$

- Let's compute:

$$k \binom{n}{k} = k \frac{n(n-1) \dots (n-k+1)}{k(k-1) \dots 1} = n \frac{(n-1) \dots (n-k+1)}{(k-1) \dots 1} = n \binom{n-1}{k-1} \quad (2)$$

Expected number of successes

- Substituting (2) to (1) yields:

$$E[S_n] = n \sum_{k=1}^n \binom{n-1}{k-1} p^k q^{n-k}$$

Observation:

The summation is very close to the binomial formula...

Expected number of successes

- Substituting (2) to (1) yields:

$$E[S_n] = n \sum_{k=1}^n \binom{n-1}{k-1} p^k q^{n-k}$$

Observation:

The summation is very close to the binomial formula.

- Now define $k - 1 = \ell$ in order to have a counter that starts from 0

$$E[S_n] = n \sum_{\ell=0}^{n-1} \binom{n-1}{\ell} p^{\ell+1} q^{n-\ell-1} = np \sum_{\ell=0}^{n-1} \binom{n-1}{\ell} p^{\ell} q^{n-1-\ell} =$$

Do we need to compute this summation analytically, or

Expected number of successes

- Substituting (2) to (1) yields:

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Observation:

The summation is very close to the binomial formula.

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$$E[S_n] = n \sum_{\ell=0}^{n-1} \binom{n-1}{\ell} p^{\ell+1} q^{n-\ell-1} = np \sum_{\ell=0}^{n-1} \binom{n-1}{\ell} p^{\ell} q^{n-1-\ell} = np \sum_{\ell=0}^{n-1} P\{S_{n-1} = \ell\} =$$

Expected number of successes

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$$E[S_n] = n \sum_{k=1}^n \binom{n-1}{k-1} p^k q^{n-k}$$

Observation:

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- Now define $k - 1 = \ell$ in order to have a counter that starts from 0

$$E[S_n] = n \sum_{\ell=0}^{n-1} \binom{n-1}{\ell} p^{\ell+1} q^{n-\ell-1} = np \sum_{\ell=0}^{n-1} \binom{n-1}{\ell} p^{\ell} q^{n-1-\ell} = np \sum_{\ell=0}^{n-1} \overbrace{P\{S_{n-1} = \ell\}}^{= 1} = np$$

(slightly different derivations from those in the lecture notes; study both)

Expected number of successes

- **Simpler and faster solution!**
- Recall that we defined $X_i \in \{0,1\}$ as the **indicator function**
 - With $P\{X_i = 1\} = p$ and $P\{X_i = 0\} = q = 1 - p$
 - Then $S_n = X_1 + X_2 + \cdots + X_n$

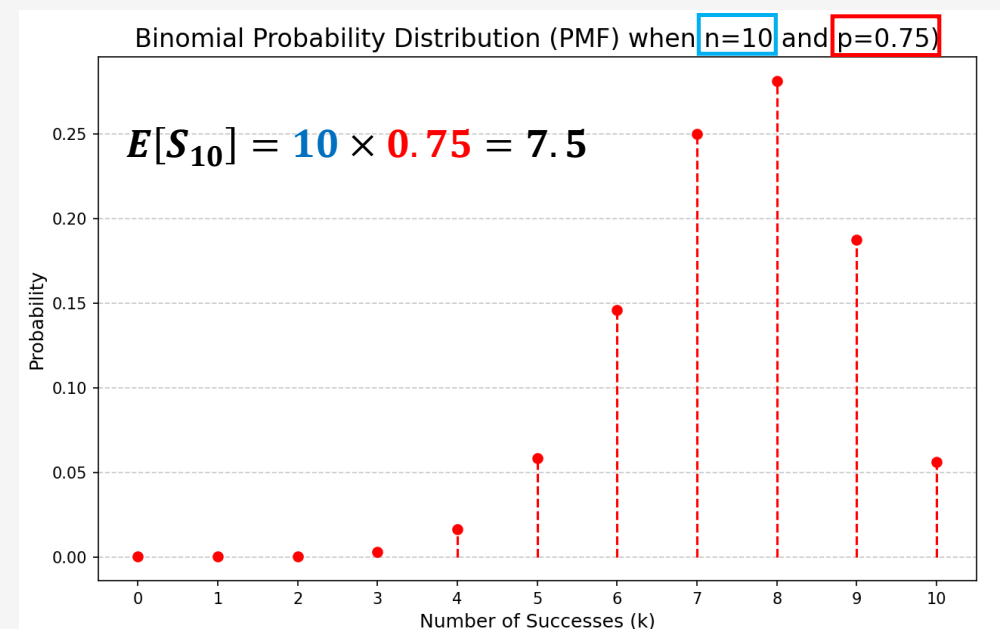
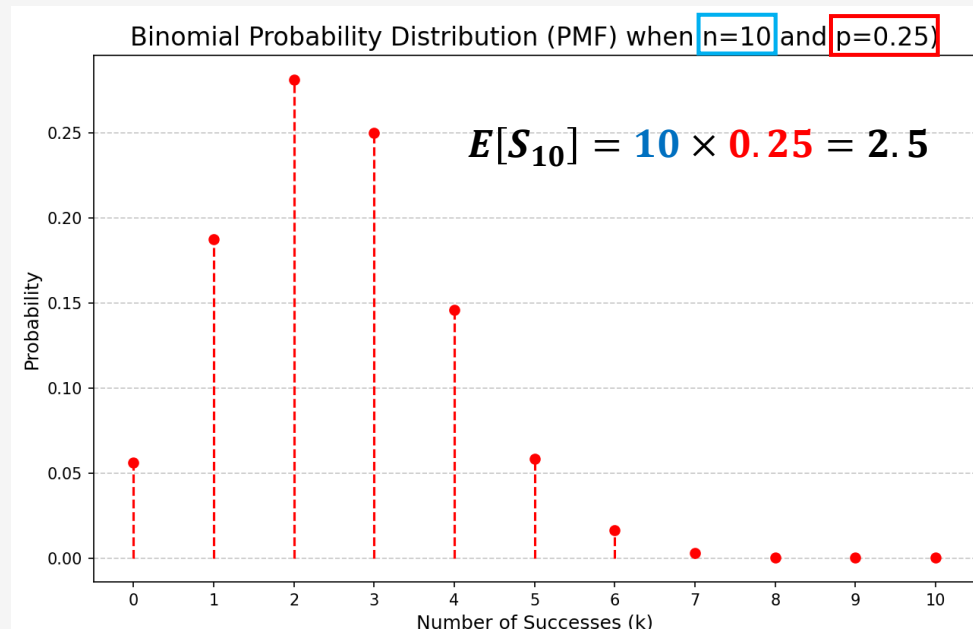
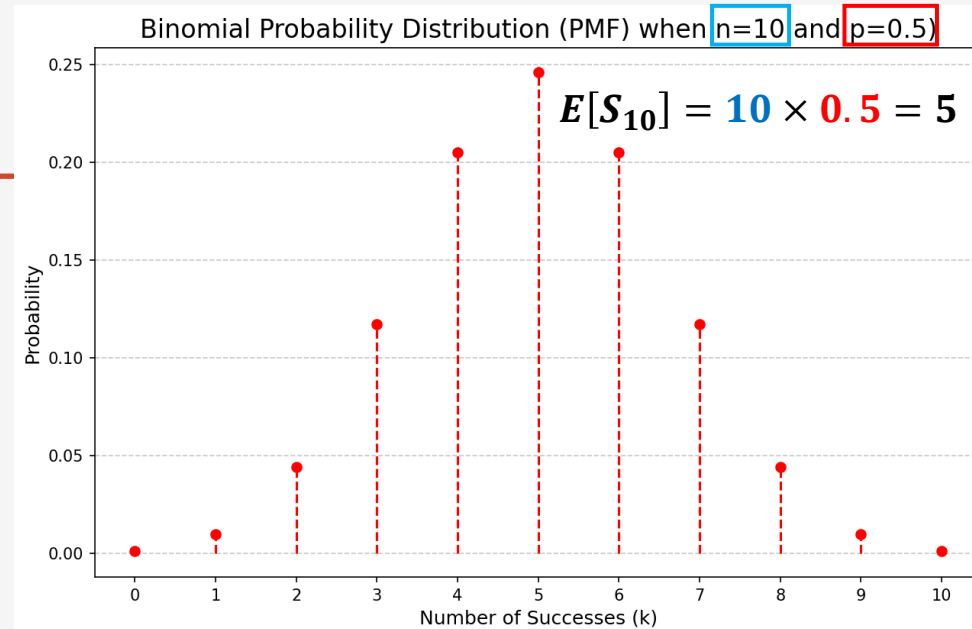
$$E[S_n] = E[X_1 + x_2 + \cdots + X_n] = E[X_1] + E[X_2] + \cdots + E[X_n] =$$

$$P[X_1 = 1] + P[X_2 = 1] + \cdots + P[X_n = 1] = np$$

$$n = 10$$

$$P\{Z = k\} = \binom{n}{k} p^k q^{n-k}$$

$$E[S_n] = np$$



Successes between trials

- **Example:**
 - Let's compute the distribution of: "# of successes between trials 10 and 21"

How can we write this in terms of S_n ??

Successes between trials

- **Example:**
 - Let's compute the distribution of: "# of successes between trials 10 and 21"
 - We are looking for the distribution of the DRV: $Z = S_{21} - S_9$

$$Z = S_{21} - S_9 = \sum_{k=1}^{21} X_k - \sum_{k=1}^9 X_k = \sum_{k=10}^{21} X_k$$

- But the DRVs $X_{10}, X_{11}, \dots, X_{21}$ have the **same distribution** as?

Successes between trials

- **Example:**
 - Let's compute the distribution of: "# of successes between trials 10 and 21"
 - We are looking for the distribution of the DRV: $Z = S_{21} - S_9$

$$Z = S_{21} - S_9 = \sum_{k=1}^{21} X_k - \sum_{k=1}^9 X_k = \sum_{k=10}^{21} X_k$$

- But the DRVs $X_{10}, X_{11}, \dots, X_{21}$ have the **same distribution** as $X_1, X_2, \dots, X_{12}$

Successes between trial

- Hence: $S_{21} - S_9 \sim S_{12} \sim \text{Binom}(12, p)$
- Now we could compute the probability to have **3 successes between the trials 10 and 21:**

$$P\{S_{21} - S_9 = 3\} = P\{S_{12} = 3\} = \binom{12}{3} p^3 q^{12-3}$$

- **Remark:** Two RVs that have the same distribution are NOT necessarily the same RVs. Hence, although $S_{21} - S_9 \sim \text{Binom}(12, p)$ and $S_{12} \sim \text{Binom}(12, p)$ they are not the same RVs! (to find 12 start from 10 and stop at 21)

Incremental successes

- **Example:**
 - What is the probability: $P\{S_3 = 2, S_7 = 3\} = ??$
 - That is, we are looking for **the probability we see 2 successes by trial 2 and 3 successes by trial 7.**
 - **Question:** are the events $\{S_3 = 2\}$ and $\{S_7 = 3\}$ **independent??**

Incremental successes

- **Example:**
 - What is the probability: $P\{S_3 = 2, S_7 = 3\} = ??$
 - That is, we are looking for **the probability we see 2 successes by trial 2 and 3 successes by trial 7.**
 - **Question:** are the events $\{S_3 = 2\}$ and $\{S_7 = 3\}$ **independent** $\rightarrow$ **NO!**
 - The number of successes we see by trial 7 depends on the trials we have seen by trial 2.

Incremental successes

- We need to find **independent events**! → *Life is better when you are independent!*
- **Observation:**
 - We can think of the event $\{S_3 = 2, S_7 = 3\}$ as the event $\{S_3 = 2, S_7 - S_3 = 1\}$
 - **Question:** are the events $\{S_3 = 2\}$ and $\{S_7 - S_3 = 1\}$ **independent??**

Incremental successes

- We need to find independent events! → Life is better when you are independent!
- **Observation:**
 - We can think of the event $\{S_3 = 2, S_7 = 3\}$ as the event $\{S_3 = 2, S_7 - S_3 = 1\}$
 - **Question:** are the events $\{S_3 = 2\}$ and $\{S_7 - S_3 = 1\}$ **independent??** → **YES!**
 - $\{S_3\} \sim \text{Binom}(3, p)$ and $\{S_7 - S_3\} \sim \text{Binom}(4, p)$

$$P\{S_3 = 2, S_7 = 3\} = P\{S_3 = 2\}P\{S_7 - S_3 = 1\} = \binom{3}{2} p^2 q^1 \binom{4}{1} p^1 q^3 = \dots$$

Summary of the notation so far

Sample space: Ω (the last letter in the Greek alphabet). Example: $\Omega = \{1,2,3,4,5,6\}$

Events: capital English letters, e.g., A,B,C,...; they are subsets of Ω .

Probability of an event $A = \{1,2\}$: $P(A) := P(\{1,2\}) := P\{1,2\}$ Caution: this is wrong: $P(1)$ or $P(1,2)$

Random variables: capital English letters, e.g., X,Y,Z, ...

Values of random variables: lowercase English letters: x, y, x, ...

An event described in terms of a random variable: $\{X = x\} \rightarrow$ it means the rv X takes value x

Probability of this event: $P(\{X = x\})$ or in a simpler form: $P\{X = x\}$; you may also see it in other books as $P(X = x)$; all notations are fine and you are free to use them in PSETs or exams. I will be using $P\{X=x\}$ but you can use any of the other two notations. **Caution:** this is wrong: $P(x)$ since x is just a value.

Practice exercises/examples

(may not be covered during lectures but they are offered for your practice)

Bayesian inference

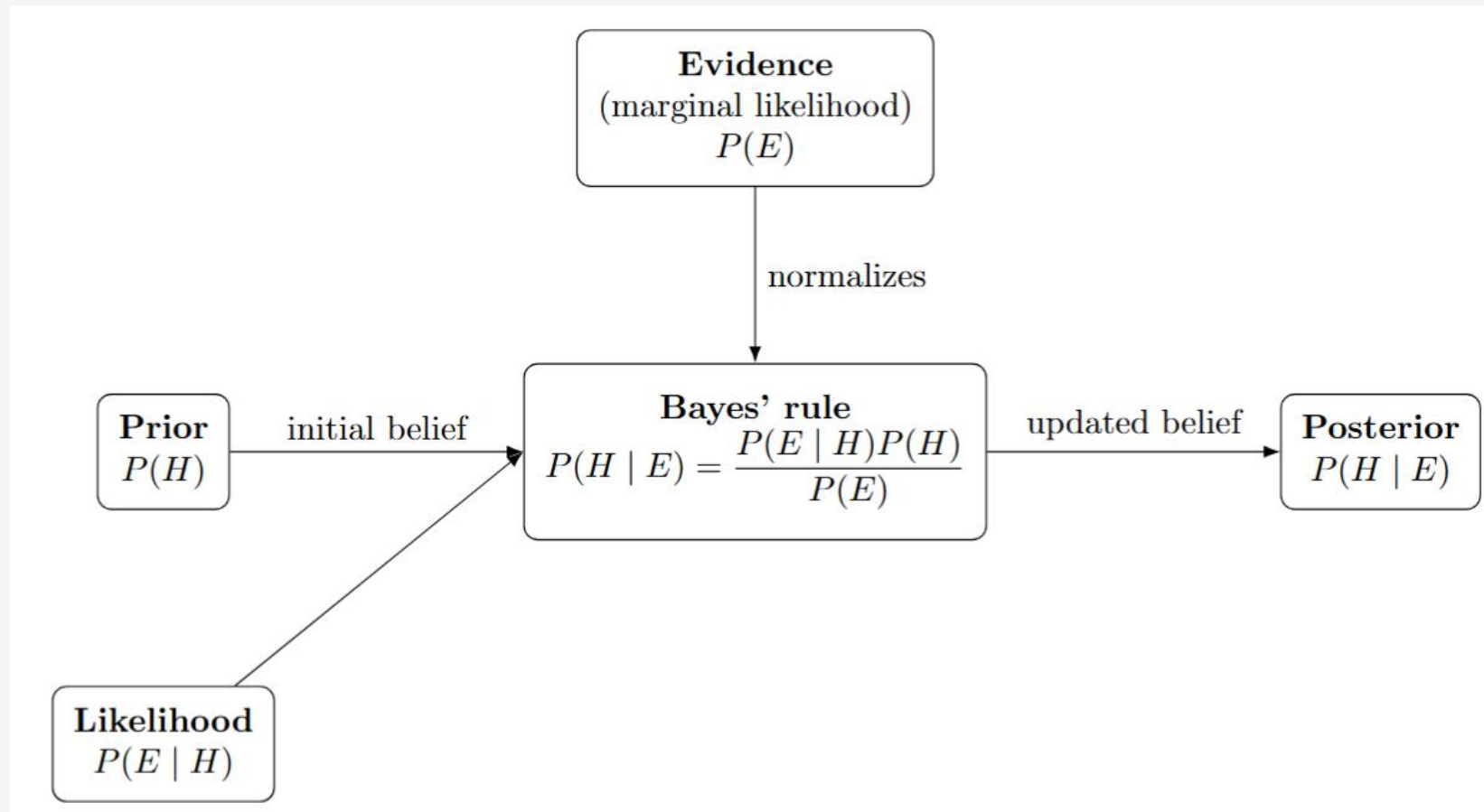
- I will discuss briefly the **mathematical framework of Bayesian inference**. This builds on the **Conditional Probability** discussed in the lecture notes, and in previous lectures.
- In Bayesian Statistics, we update our beliefs about the hypothesis based on new data/information. The update is governed by Bayes Formula:

$$P(H|E) = \frac{P(H \cap E)}{P(E)} = \frac{P(E|H) P(H)}{P(E)}$$

where the event H is called the **hypothesis**, and the event E the **evidence**.

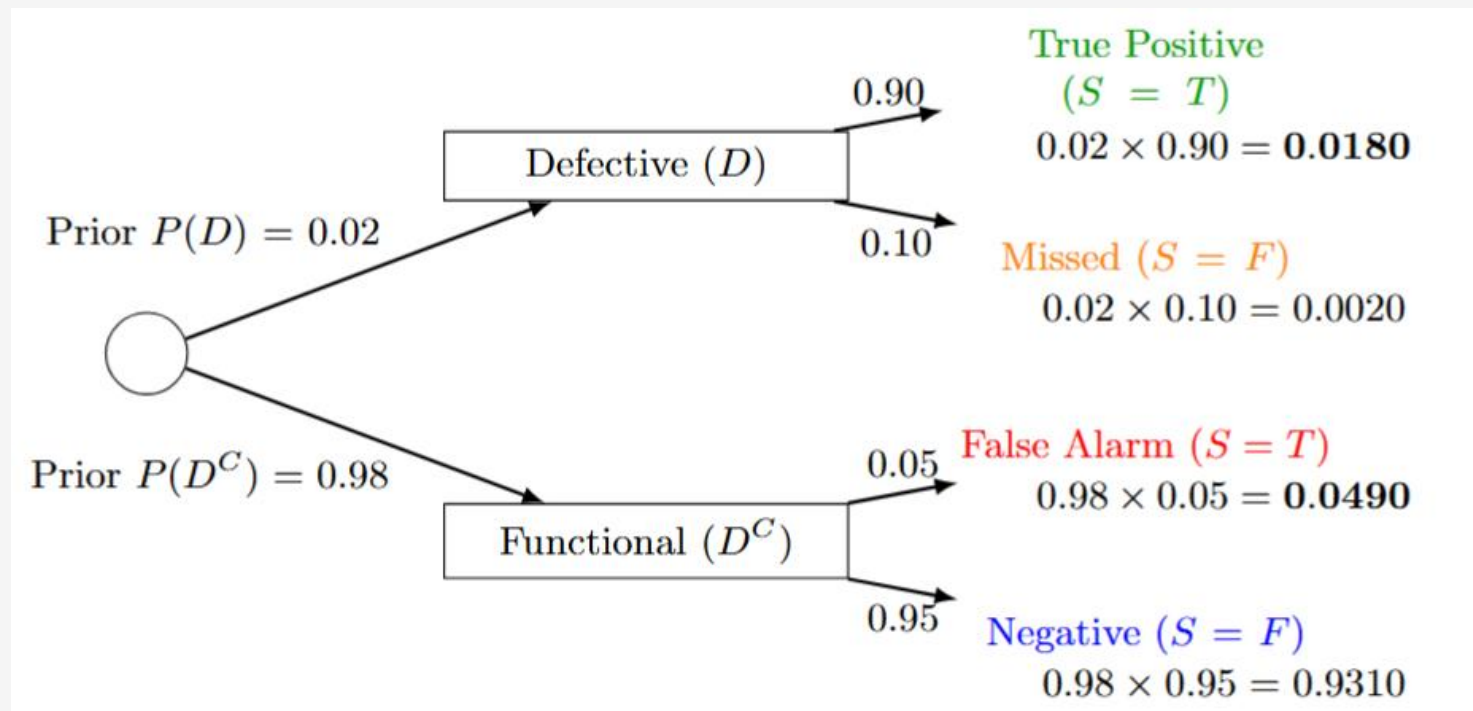
- $P(H)$ is called the **prior probability** and represents the confidence of our belief in the **hypothesis** H , before we observe the **evidence** E . In a way $P(H)$ is our basic knowledge.
- $P(E|H)$ is our confidence we have that the evidence E will be observed **given that** the hypothesis H is true.
- $P(E)$ is the probability of **observing the data/information under all possible hypotheses**. Note that $P(E) = P(E|H) P(H) + P(E|H^c) P(H^c)$
- $P(H|E)$ is the **posterior probability** which represents the probability of the hypothesis H after the evidence E has been taken into account.

Bayesian inference



Bayesian inference

- Let's see an example in Quality Control
- Suppose we work in a light bulb factory, and we want to find the probability that a bulb is **defective** (D) based on positive test performed by a **sensor** (S).
- From **historical data** we know that only 2% of the bulbs are defective: $P(D) = 0.02$ (and $P(D^c) = 0.98$). This is the **prior probability** we have been given or computed.
- The sensor can declare a defect 90% of the times if a defect exists ($P(S|D) = 0.90$), and 5% of the times when there is no defect ($P(S|D^c) = 0.05$; false positive rate or false alarm). This is the **evidence** (or **likelihoods**) we have.
- The **total evidence (or marginal likelihood)** is $P(S) = P(S|D)P(D) + P(S|D^c)P(D^c) = 0.90 \times 0.02 + 0.05 \times 0.98 = 0.067$
- The **posterior probability** is: $P(D|S) = \frac{P(S|D)P(D)}{P(S|D)P(D) + P(S|D^c)P(D^c)} = \frac{0.90 \times 0.02}{0.067} \approx 0.2687$.
- **Conclusion:** Even though the sensor indicated the bulb is defective ($S = \text{True}$), the posterior probability is only 26.87%. This occurs because the prior probability of a defect ($P(D) = 0.02$) is smaller than the false alarm rate ($P(S|D^c) = 0.05$).



Conditional expectations

- **Exercise:** An investment company is currently analyzing the performance of its flagship Tech-heavy ETF in anticipation of the Fed's upcoming interest rate decision. They have come up with the following analysis:
 - The ETF's return is a DRV (X) and can take the values: -10% , 0% or $+10\%$.
 - The Fed's rate change is also a DRV (Y) and can take the values: 0 , 25 and 50 (basis points)
 - The joint probability distribution of these DRVs is given by the table

$P(X=x, Y=y)$	0	25	50
-10%	0.05	0.10	0.20
0%	0.10	0.15	0.05
+10%	0.25	0.05	0.05

Conditional expectations

- **Q1:** Compute the marginal probability distribution of the Fed's decision. Which rate is the most likely to happen?
- **Q2:** If the Fed announces a 50 basis points increase, calculate the conditional expected return of the ETF.
- **Q3:** Calculate the unconditional expected return of the ETF ($E[X]$).
- **Q4:** The company's strategy is to only hold assets with an expected return rate $\geq 0\%$.
 - **a.** Would the company have held the ETF before the Fed announces its new rates?
 - **b.** Based on your result in Q2, should the company sell or keep the ETF after the Fed makes its announcement of 50 basis points increase?